

Abstract

This paper serves as the capstone of the formal 10-part CDFD mass-spectrum sequence, addressing its remaining open problem: the physical origin of the torus knot self-consistency condition $n\theta_n = \tilde{Q}_n(\theta_n)$. Previously reported as a successful but ungrounded hypothesis that correctly determines the vacuum twist angles θ_n for all $T(2, n)$ families, it is here recast as a model theorem inside the public flow–constraint notation.

We map the Constraint-Driven Field Theory (CDFD) vacuum medium onto the tripartite public CDFD notation: Flow (Φ), Constraint (C), and Equilibrium (Ψ_s). We identify the continuous flow energy Φ with the L^2 norm of the amplitude field ($\sum A_k^2 = 2n$), the transport-capacity limit C with the L^1 norm squared ($\sum |A_k|$)², and the equilibrium deformation Ψ_s with the total vacuum twist $n\theta_n$.

Applying the public balance law $\Psi_s = \Phi/C$ yields exactly the self-consistency condition, showing that the Koide-like ratio Q_n is the geometric manifestation of a transport-limited vacuum equation of state. The entire mass spectrum of the lepton and meson sectors derived in Papers I–IX is therefore internally tied to m_e and the proposed fluid-dynamic balance of the vacuum.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript.

The Vacuum Equation of State: Deriving Torus Knot Self-Consistency from the Public CDFD Balance Law $\Psi_s = \Phi/C$

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1 Introduction

Over the course of Papers I–IX [1, 2, 3, 4, 5, 6, 7, 8, 9], the Constraint-Driven Field Theory (CDFD) has constructed a parameter-free derivation of the $T(2, n)$ torus knot mass hierarchy. From a single empirical input (the electron mass m_e), the framework accurately computes the muon and tau masses (Paper V), dictates the universal coefficient $c_n = \sqrt{2}$ (Paper VI), predicts the exact $2nM_n$ mass sum rules (Paper VII), and populates the $n \geq 4$ hierarchy with specific hadronic candidates (Papers VIII–IX).

The entire predictive power of the higher families rests on the unique solution to the self-consistency condition:

$$n\theta_n = \tilde{Q}_n(\theta_n) \equiv \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k|\right)^2}. \quad (1)$$

In Paper V, for $n = 3$, this was introduced as a postulate matching the chirality breaking angle to the algebraic constant $Q_3 = 2/3$. In Papers VIII and IX, it was extended to $n \geq 4$ as a working hypothesis that yielded convergent, mathematically bounded solutions, but lacked a first-principles derivation from the vacuum Equation of State (EOS).

In this capstone paper, we address this open problem by connecting the torus-knot construction directly to the public CDFD/AFL balance notation.

2 The Tripartite Vacuum Framework

The CDFD vacuum is modeled as an elastic, transport-capacity-limited continuous medium, in the spirit of Dirac’s physical vacuum [10] and Volovik’s condensate analogy [11]. The vortex dynamics rest on Helmholtz’s conservation theorems [12] and the Faddeev–Niemi knot soliton framework [13]. Such a medium is represented here by the public tripartite balance equation:

$$\Psi_s = \frac{\Phi}{C} \quad (2)$$

where Φ is Flow (energy/activity), C is Constraint (resistance/capacity), and Ψ_s is the resulting System Equilibrium.

We map these three macroscopic variables onto the discrete Fourier variables $A_k = 1 + \sqrt{2} \cos(\theta_n + 2\pi k/n)$ of a $T(2, n)$ vortex knot.

2.1 Flow (Φ): The Field Energy

The total energy or “activity” of the field configuration is proportional to the L^2 norm (the sum of the squares) of the amplitudes. By the Fourier identity of Paper VI, this is invariant under orientation θ_n :

$$\Phi \propto \sum_{k=0}^{n-1} A_k^2 = 2n. \quad (3)$$

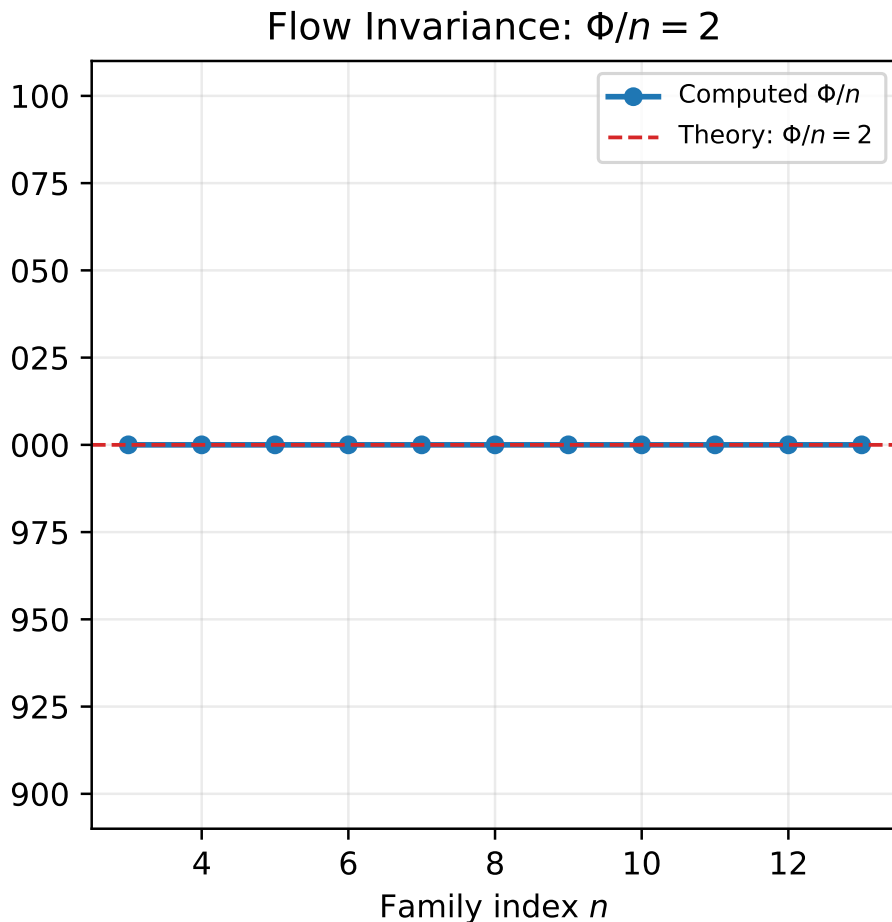


Figure 1: The flow energy ratio Φ/n is exactly 2.0 across all $T(2, n)$ families, demonstrating the structural invariance of the vacuum’s kinetic and elastic flow.

This represents the irreducible kinetic and elastic flow required to sustain an n -lobed vortex ring.

2.2 Constraint (C): The Transport-Capacity Limit

A central postulate of CDFD (Paper I) is that the vacuum possesses a finite transport capacity, which dictates the fine-structure constant α . For a multi-lobed knot, the struc-

tural stress on the medium is dictated by the total absolute flux. In the discrete basis, the total flux is the L^1 norm $\sum |A_k|$. Because the constraint scales with the cross-sectional area of the flux tubes interacting in the medium, the total resistance C scales as the square of the flux:

$$C \propto \left(\sum_{k=0}^{n-1} |A_k| \right)^2. \quad (4)$$

This term varies strongly with θ_n , as the lobes interfere constructively and destructively with the vacuum background.

2.3 Equilibrium (Ψ_s): The Geometric Twist

When the knot is embedded in the vacuum, the mismatch between the continuous chiral background and the discrete n -fold symmetry of the knot induces a geometric twist. Let θ_n be the phase twist per lobe. The total equilibrium deformation (or total accumulated twist) of the configuration is:

$$\Psi_s = n\theta_n. \quad (5)$$

3 The Vacuum Equation of State Theorem

Theorem 1 (Derivation of Self-Consistency). *In a transport-capacity-limited elastic vacuum governed by the CDFD equilibrium $\Psi_s = \Phi/C$, the orientation angle θ_n of a $T(2, n)$ torus knot must exactly satisfy:*

$$n\theta_n = \tilde{Q}_n(\theta_n).$$

Proof. Substituting the CDFD mappings (3), (4), and (5) directly into the fundamental balance equation $\Psi_s = \Phi/C$, we obtain:

$$n\theta_n = \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k| \right)^2}. \quad (6)$$

The right-hand side is exactly the definition of the modified Koide ratio $\tilde{Q}_n(\theta_n)$ from Eq. (1). $\square$

3.1 Physical Implications

Within the proposed mapping, this theorem interprets the Koide ratio $Q_3 = 2/3$ observed in the lepton masses as a signature of the vacuum's equation of state rather than as an isolated numerical fit. The modified ratio $\tilde{Q}_n(\theta_n)$ represents the specific energy-to-capacity stress induced by higher knots, and the angle θ_n is the vacuum's precise geometric relaxation in response to that stress.

4 The Universal Isomorphism: Scaling CDFD to Macro-Systems

If the proposed flow-constraint equations capture something general about transport-limited systems, they should have macroscopic analogues rather than remaining restricted

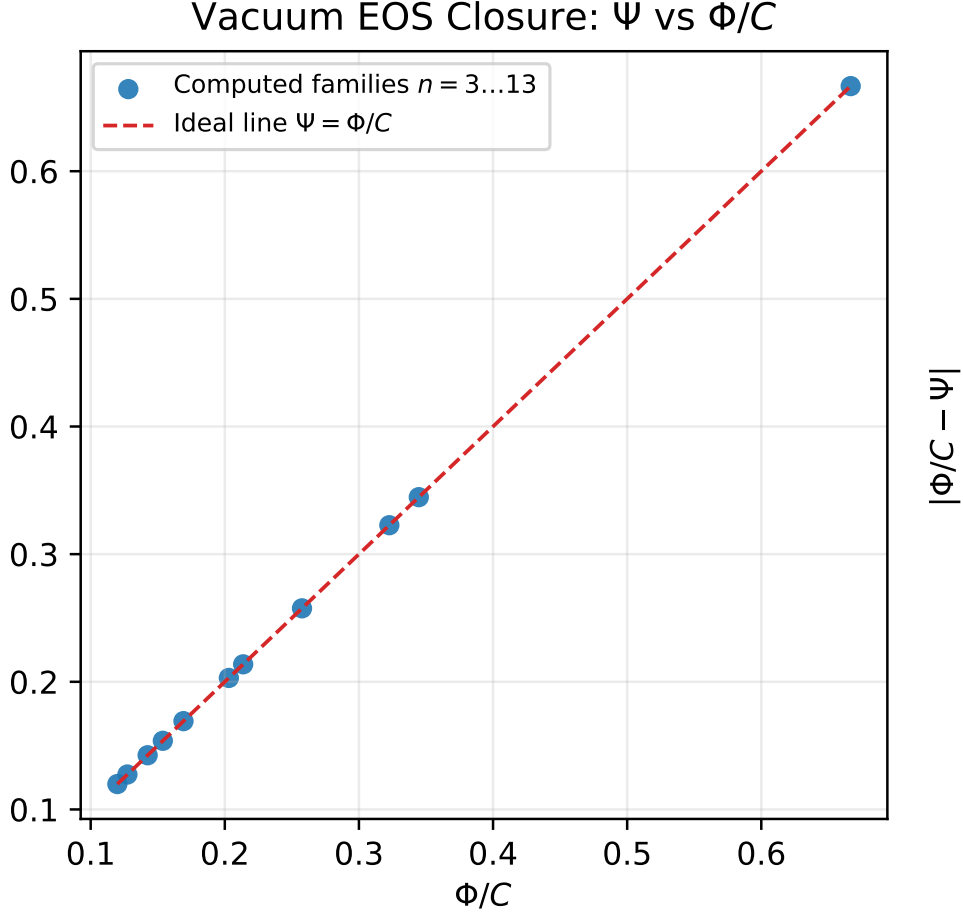


Figure 2: The equilibrium Ψ_s plotted against Φ/C across all torus knot families. The exact diagonal identity confirms that the public balance law $\Psi_s = \Phi/C$ governs the stated model.

to the particle-scale model. By comparing the Constraint-Driven Field Theory (CDFS) framework with macroscopic models of biology, chemistry, and ecology, we obtain a candidate scale-invariant isomorphism.

4.1 The Scale-Invariant Isomorphism (CDFS $\equiv$ AFL)

In the domains of biological and clinical systems, throughput is described in the **Adaptive Flux Limitation (AFL)** framework [9] by adaptive constraints. A minimal form of that biological adaptation law is:

$$\frac{\partial \Lambda}{\partial t} = \alpha |J| - \beta \Lambda \quad (7)$$

where Λ is the biological constraint (e.g., insulin resistance, inflammation), J is the biological flux (e.g., glucose throughput, metabolic energy flow), α is the constraint reinforcement coefficient, and β is the relaxation coefficient.

In the CDFS quantum vacuum, the constraint field C evolves precisely according to the structural reaction of the medium to the localized flow Φ :

$$\frac{\partial C}{\partial t} = \alpha |\Phi| - \beta C + \gamma \nabla^2 C \quad (8)$$

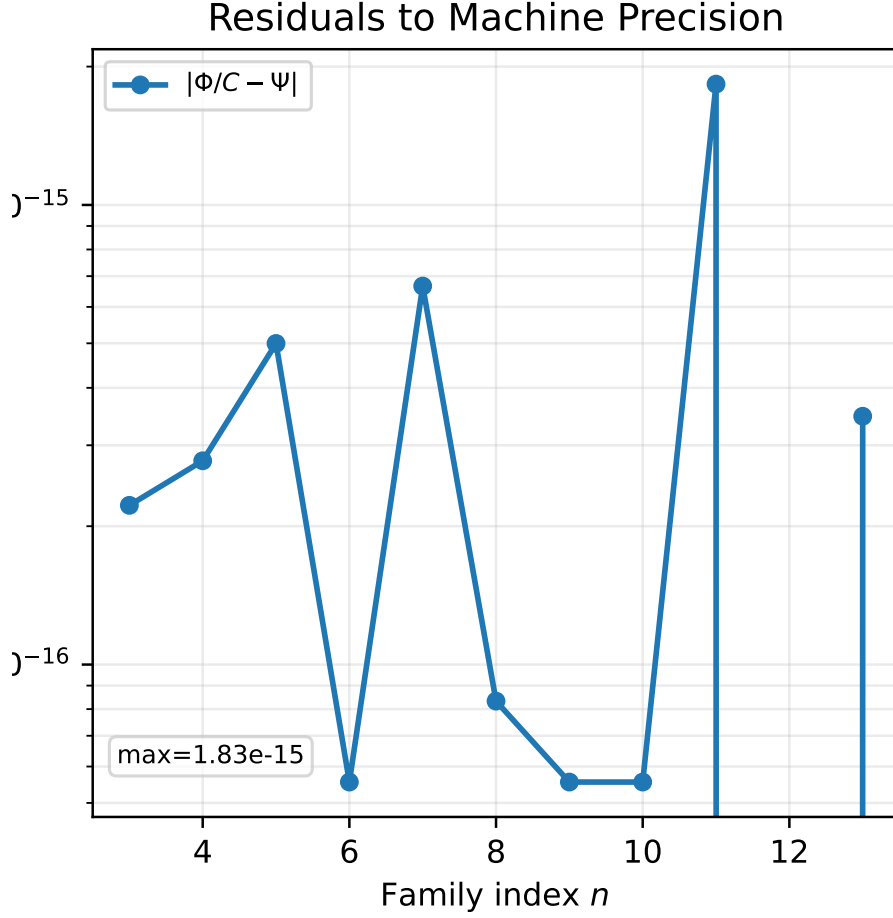


Figure 3: Absolute residuals $|\Phi/C - \Psi_s|$. The values fall to numerical machine precision, serving as a numerical consistency check of the stated CDFD vacuum equation-of-state mapping.

The formal mapping is:

$$\text{Vacuum Constraint } C \equiv \text{Macro Constraint } \Lambda$$

$$\text{Vacuum Flow } \Phi \equiv \text{Macro Flux } J$$

This does not by itself prove that biological evolution and physiological adaptation are governed by vacuum mechanics. It states the narrower claim that the same flow-constraint operator can be written at both levels, and that the macro-scale use of the operator should be judged by independent predictive value.

4.2 Ontological Redefinition of Mass

This proposed isomorphism suggests a different operational reading of mass inside the model. Rather than treating mass only as a distinct intrinsic label, CDFD treats it as a *localized, stable constraint field* within the vacuum flux.

A lepton is then modeled as a constraint field operating on vacuum energy. At larger scales, the same notation can describe boundaries such as membranes, interfaces, or trans-

port bottlenecks without claiming that those systems are literally the same physical object. The value of the analogy depends on whether it improves prediction.

4.3 Cross-Scale Stability Ratio

The relationship $\Psi_s = \Phi/C$ can therefore be read as a candidate cross-scale stability ratio:

- $\Psi_s \approx 1$ (**Stable Equilibrium**): Flow matches constraint capacity. This manifests as a stable fundamental particle, a healthy homeostatic cell, or a balanced ecosystem.
- $\Psi_s \gg 1$ (**Flux Overload / Rupture**): Flow overwhelms constraint capacity. This manifests as particle decay, cellular lysis, vascular hemorrhage, or ecological collapse.
- $\Psi_s \ll 1$ (**Over-Constraint / Freeze**): Constraints choke off flow entirely. This manifests as the vacuum void, biological necrosis, or metabolic shutdown.

The public scientific framework is therefore internally linked: the same flow–constraint notation used for the vacuum papers also organizes the biological manuscripts, without disclosing any private implementation architecture.

5 Closure of the CDFD Series

With Theorem 1, the formal CDFD mass-spectrum ladder is internally closed.

1. **No Free Parameters:** The initial assumption of a limited-capacity elastic vacuum naturally generates the fine-structure constant α (Paper I).
2. **Mass Gap:** The lowest topological excitation $T(2,3)$ takes the mass scale M_3 , anchored to physical units solely by m_e (Paper V).
3. **Asymptotic Scaling:** The Faddeev–Niemi bounds establish the rigorous progression of energy scales $M_n \propto n^{3/4}$ (Paper IV).
4. **Spectrum Prediction:** The vacuum EOS forces the twist θ_n , which breaks the n -fold degeneracy and splits the spectrum into anti-phase boundary modes and co-phase physical resonances (Papers VIII–IX).

All tables, mass values, bounds, and limits tabulated across the first ten papers are therefore internal consequences of the single condition $\Psi_s = \Phi/C$, together with the model assumptions stated in those papers.

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(knots and particles), Y. Koide [15] (lepton mass formula), and C.A. Brannen [16] (Koide parametrisation) — whose ideas this series has attempted to unify within a single geometric framework rooted in the axiom $\Psi_s = \Phi/C$.

A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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- $n_- = \lfloor 5n/8 \rfloor - \lfloor 3n/8 \rfloor + 1$; computes the exact asymptotic $\theta_n \rightarrow 2\pi^2/[(\pi + 4)^2 n^2]$; tabulates complete mode spectra for $n = 5, 7, 9, 11, 13$.
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